Time Series Forecasting of Wikipedia Web Traffic

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May 19, 2025

1 K-means

Overview

This section describes the approach used to forecast time series for web traffic data by using a combination of unsupervised learning and an adaptive statistical method. Here, we are using K-means with an adaptive moving average. So, our aim is to group/cluster similar page-based viewing patterns and apply a dynamic moving average to predict future values.

Data Preparation

Here, a **preprocessed** dataset done by our teammates is used here. It consists of page names and their daily view counts for Wikipedia articles. Each row represents page traffic and time series for each page.

1. **Loading the dataset:** reading and loading the CSV file is done by `using _read_csv` from pandas. The dataset's first column is page names, and the remaining columns contain daily time series.

2. **Datatype conversion:** converting all time series values to float type. This will avoid datatype-related problems.

3. **Standardization:** each time series is standardized by utilizing this formula:

$$X_{\text{scaled}} = \frac{X - \mu}{\sigma}$$

   Where,

   - μ is the *mean* of rows;
   - σ is the *standard deviation* of rows;
   - If $\sigma = 0$, then we'll replace it with 1.

   This will change each series to zero mean and unit variance. **It is crucial for clustering** to treat every series equally, regardless of its original size.

4. **Return values:**
   - page names for identifying;
   - standardized traffic data;
   - mean and standard deviation for reversing standardization;
   - original data frame.

Feature engineering

To have meaningful clustering of time series, a set of summary statistics has been taken from standardized time series.

1. **Mean:** it's the mean value of every series for all days.
2. **Standard deviation:**
   - calculates fluctuations in each series;
   - used for dynamic forecasting window size.
3. **Peak (maximum) and lowest (minimum):** finds out the High and Low values.
4. **30-Day average:** mean of the last 30 days.
5. **Spike count:** count of how many days the traffic spiked above the threshold.

   the spike threshold is defined as:

   $$\text{threshold} = \mu + 2 \times \sigma$$

   the spike count is then computed as:

   $$\text{Spikecount} = I(x_1 > \text{threshold}) + I(x_2 > \text{threshold}) + \cdots + I(x_T > \text{threshold})$$

   where 1 is added if the condition is true, else 0.

Model architecture and configuration

Unsupervised learning with statistical forecasting

KMeans clustering

- number of clusters: 12;
- random seed: 42;
- random sampling: 10,000 time series;
- `KMeans` applied to samples;
- **Features:**

  mean, standard deviation, maximum, minimum, average of last 30 days, and spike count;
- **Standardization** using `StandardScaler`;
- label all the data using KNN, $k = 1$.

```
Cluster 0: 20450 series, Adaptive MA window = 30 (raw std = 2285.8442)
Cluster 1: 18650 series, Adaptive MA window = 30 (raw std = 2671.9322)
Cluster 2: 19 series, Adaptive MA window = 30 (raw std = 208506.8920)
Cluster 3: 16515 series, Adaptive MA window = 30 (raw std = 3869.9409)
Cluster 4: 22302 series, Adaptive MA window = 30 (raw std = 2547.7134)
Cluster 5: 16832 series, Adaptive MA window = 30 (raw std = 3085.8025)
Cluster 6: 8509 series, Adaptive MA window = 30 (raw std = 4755.8442)
Cluster 7: 25455 series, Adaptive MA window = 30 (raw std = 3153.1954)
Cluster 8: 1210 series, Adaptive MA window = 30 (raw std = 12616.0437)
Cluster 9: 521 series, Adaptive MA window = 30 (raw std = 44725.5346)
Cluster 10: 21 series, Adaptive MA window = 30 (raw std = 681469.4419)
Cluster 11: 13 series, Adaptive MA window = 30 (raw std = 7380091.0700)
```

Figure 1: clusters and their rows

Adaptive moving average

- **split:** Train 90% and Test 10%;
- forecast the window size for each cluster using the standard deviation

$$\text{window} = \begin{cases} 3, & \text{if } \sigma \times 0.1 < 3 \\ \sigma \times 0.1, & \text{if } 3 \leq \sigma \times 0.1 \leq 30 \\ 30, & \text{if } \sigma \times 0.1 > 30 \end{cases}$$

where σ is the standard deviation, and the range ensures the result stays within [3, 30];

- **stepwise Growing Window:**
  calculate the average of the past window value for each step;
- forecasting is done on 10% of the time series (the **Forecast Horizon**);
- forecast is dynamic **w.r.t.** volatility.

Evaluation

Evaluation of predictions by using three metrics.

Mean squared error (MSE)

$$\text{MSE} = \frac{1}{N}\left[(y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 + \cdots + (y_N - \hat{y}_N)^2\right]$$

Root mean squared error (RMSE)

$$\text{RMSE} = \sqrt{\frac{1}{N}\left[(y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 + \cdots + (y_N - \hat{y}_N)^2\right]}$$

Mean absolute error (MAE)

$$\text{MAE} = \frac{1}{N}\left[|y_1 - \hat{y}_1| + |y_2 - \hat{y}_2| + \cdots + |y_N - \hat{y}_N|\right]$$

- y_i: actual value at time i;
- $\hat{y}_i$: predicted value at time i;
- N: total forecasted points.

```
[Overall] MSE: 0.5329 | MAE: 0.3265 | RMSE: 0.7300
[Cluster 0] MSE: 0.6349 | MAE: 0.3158 | RMSE: 0.7968
[Cluster 1] MSE: 0.5690 | MAE: 0.4280 | RMSE: 0.7543
[Cluster 2] MSE: 0.0003 | MAE: 0.0079 | RMSE: 0.0167
[Cluster 3] MSE: 0.2965 | MAE: 0.1324 | RMSE: 0.5445
[Cluster 4] MSE: 0.6075 | MAE: 0.3710 | RMSE: 0.7794
[Cluster 5] MSE: 0.4532 | MAE: 0.2101 | RMSE: 0.6732
[Cluster 6] MSE: 0.4682 | MAE: 0.3734 | RMSE: 0.6843
[Cluster 7] MSE: 0.6013 | MAE: 0.4178 | RMSE: 0.7754
[Cluster 8] MSE: 0.2389 | MAE: 0.2331 | RMSE: 0.4888
[Cluster 9] MSE: 0.5508 | MAE: 0.1340 | RMSE: 0.7421
[Cluster 10] MSE: 0.2093 | MAE: 0.3033 | RMSE: 0.4575
[Cluster 11] MSE: 0.0592 | MAE: 0.0940 | RMSE: 0.2433
```

Figure 2: model evaluation overview

Visualization

Plotting different rows to understand how the model behaves.

- **Per cluster:**
  selecting two random series from the `KMeans`:
  - train Values;
  - actual Values;
  - predicted Values.
- **Original scale:**
  change the forecast values by reversing using the mean and standard deviation.

this can be used to learn about accuracy and prediction stability.

per cluster:

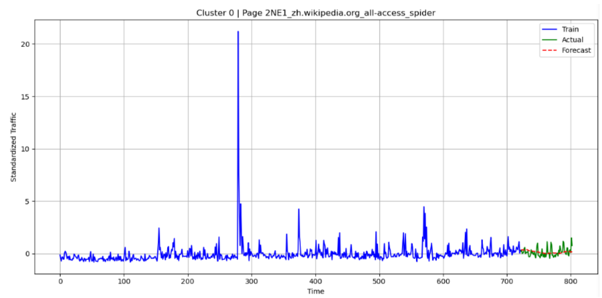

Figure 3: Random page (2NE1zh) from cluster 0

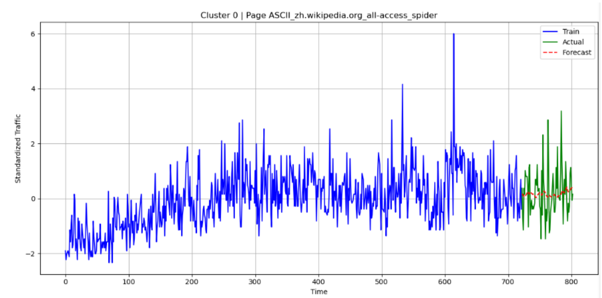

Figure 4: Random page (ASCIIzh) from cluster 0

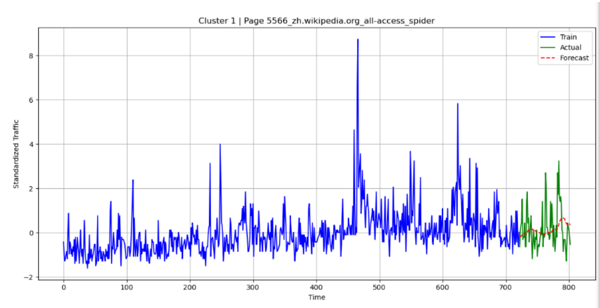

Figure 5: Random page (5566zh) from cluster 1

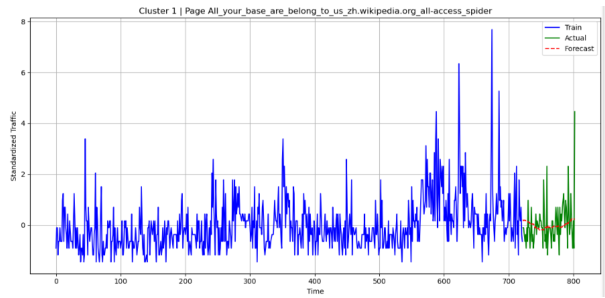

Figure 6: Random page (ALLYOURzh) from cluster 1

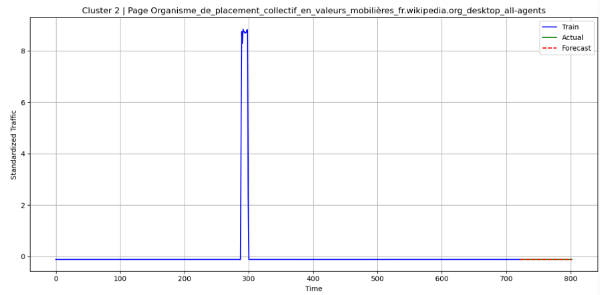

Figure 7: Random page (ORGANISMEzh) from cluster 2

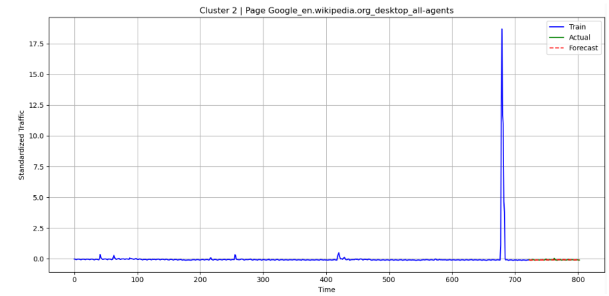

Figure 8: Random page (GOOGLEzh) from cluster 2

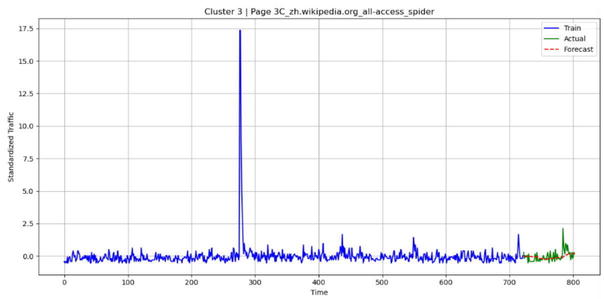

Figure 9: Random page (3Czh) from cluster 3

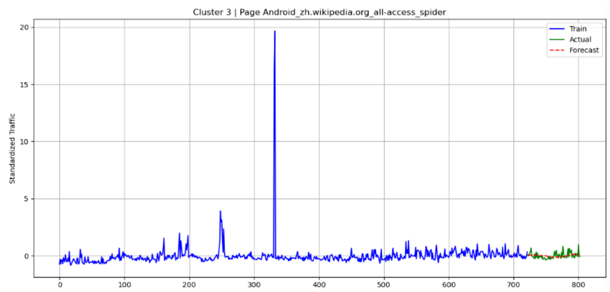

Figure 10: Random page (Androidzh) from cluster 3

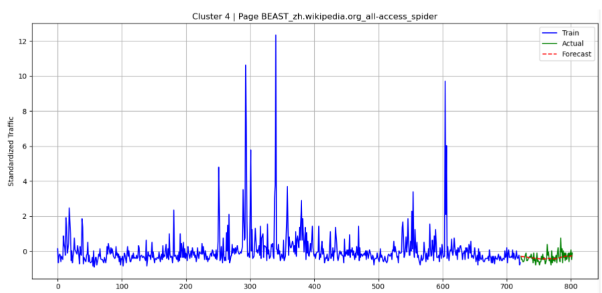

Figure 11: Random page (BEASTzh) from cluster 4

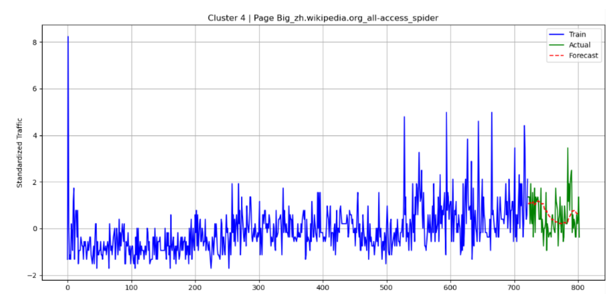

Figure 12: Random page (BIGzh) from cluster 4

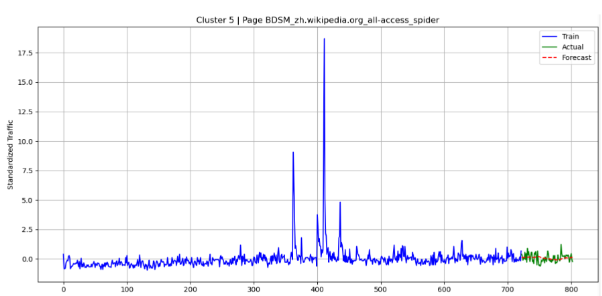

Figure 13: Random page (BDSMzh) from cluster 5

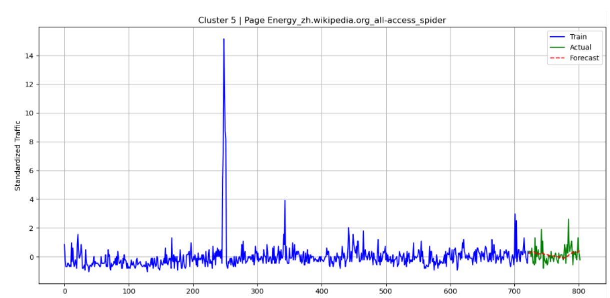

Figure 14: Random page (Energyzh) from cluster 5

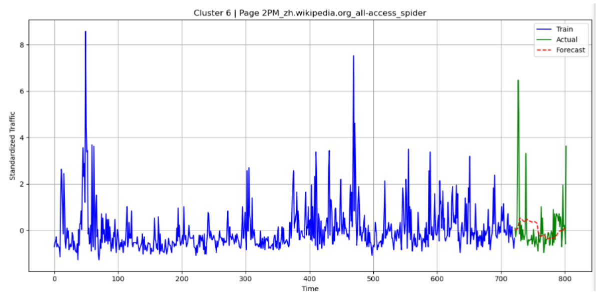

Figure 15: Random page (2PMzh) from cluster 6

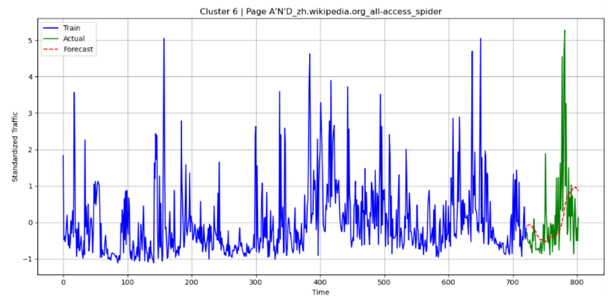

Figure 16: Random page (ANDzh) from cluster 6

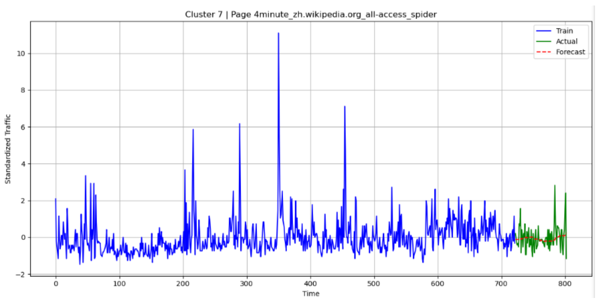

Figure 17: Random page (4minutezh) from cluster 7

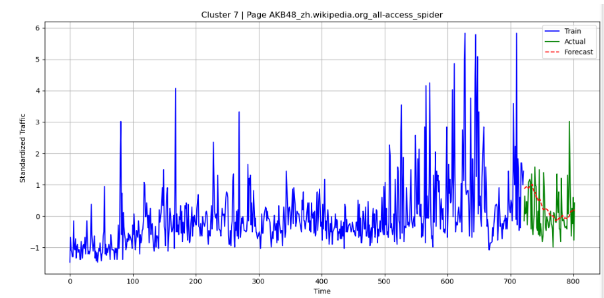

Figure 18: Random page (AKB48zh) from cluster 7

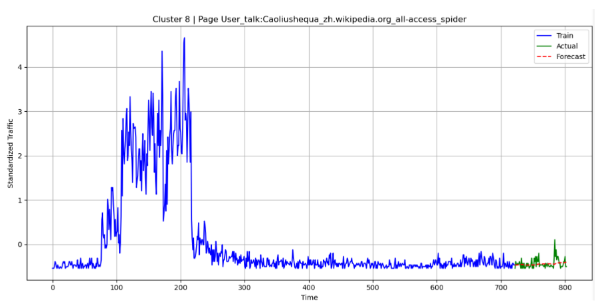

Figure 19: Random page (USERzh) from cluster 8

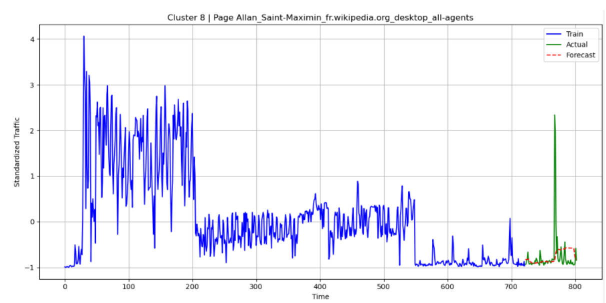

Figure 20: Random page (ALLENzh) from cluster 8

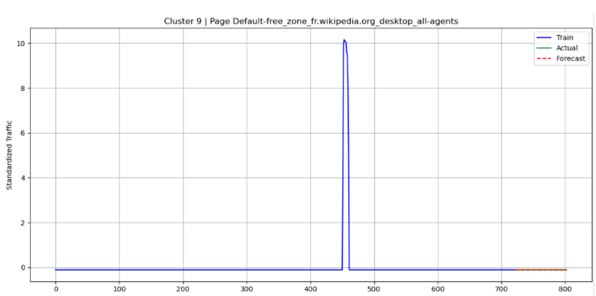

Figure 21: Random page (DEFAULTzh) from cluster 9

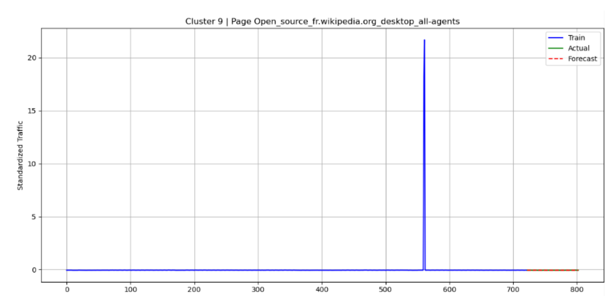

Figure 22: Random page (OPENzh) from cluster 9

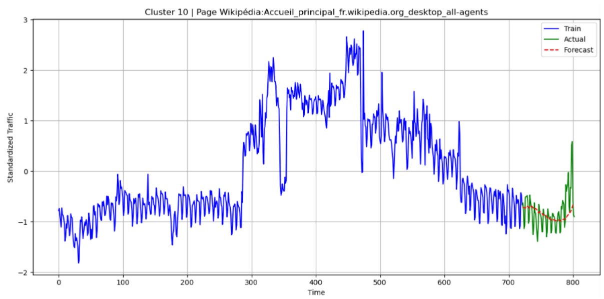

Figure 23: Random page (WIKIzh) from cluster 10

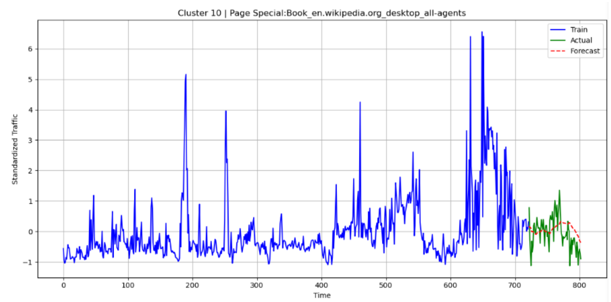

Figure 24: Random page (SPECIALzh) from cluster 10

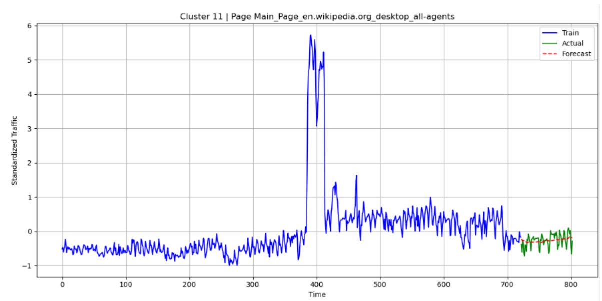

Figure 25: Random page (MAINzh) from cluster 11

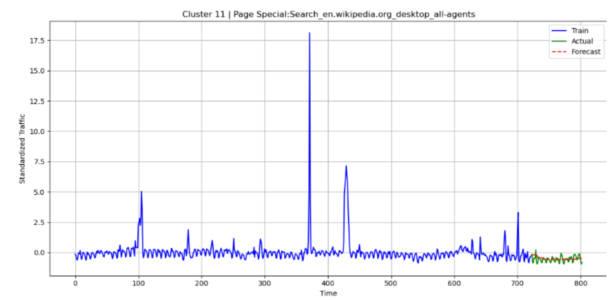

Figure 26: Random page (SEARCHzh) from cluster 11

original scale:

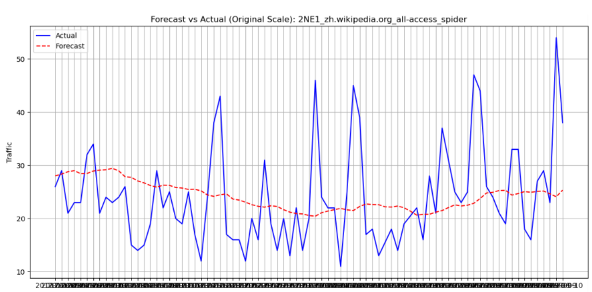

Figure 27: Random page (2NE1zh) of Forecast V Actual

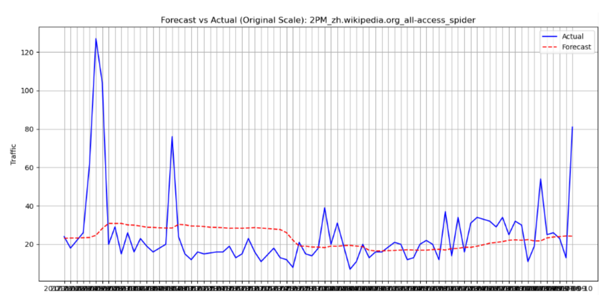

Figure 28: Random page (2PMzh) of Forecast V Actual

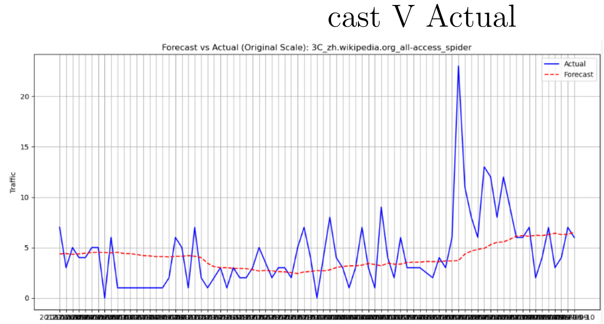

Figure 29: Random page (3Czh) of Forecast V Actual

Results

Clustering quality: calculating inertia to determine the cluster quality. Inertia measures the sum of squared distances from each point to the main cluster.

lower inertia of **624.70**, which means the time series is tightly grouped.

Forecast accuracy: final values are obtained by using error metrics

overall metrics summary

metric	**values**
MSE	*0.5329*
MAE	*0.3265*
RMSE	*0.7300*

the results above indicate that:

- the model achieved **decent accuracy**.
- **cluster-specific windowing** aided adjusted to volatility in different series.